

IO Unit

Test Report

Comprehensive Verification of PolyFlop8 IO Unit

Document ID: TC-IO-001
Revision: 1.0
Date: January 5, 2026
Test Level: Unit/Block Level
Test Engineer: Artem Horiunov
Status: **PASS**

1 TEST OVERVIEW

1.1 Unit Under Test (UUT)

Component	Input/Output Unit for PolyFlop8 Processor
VHDL Entity	IO_Unit
Source File	IO_Unit.vhd, tb_IO_Unit.vhd
Test Environment	ModelSim/QuestaSim 2023.4, VHDL Testbench
Clock Frequency	100MHz (10ns period)
Data Width	8-bit

1.2 Scope

This test report documents the verification of the IO Unit for the PolyFlop8 Processor. The IO Unit is responsible for handling input and output operations, interfacing with external devices, and managing data transfer between the CPU and physical pins.

1.3 Test Objectives

- Verify correct operation of the IO Unit during reset
- Validate write and read functionality for Port A and Port B
- Confirm data retention and address isolation
- Ensure 100% code coverage of IO Unit behavioral description

2 REQUIREMENTS COVERAGE

Req ID	Requirement Description	Test Case ID	Status
TR-IO-01	Verify write to Port A	TC-IO-001-02	PASS
TR-IO-02	Verify data retention	TC-IO-001-03	PASS
TR-IO-03	Verify read from Port B	TC-IO-001-04	PASS
TR-IO-04	Verify data integrity during hold	TC-IO-001-03	PASS
TR-IO-05	Verify reset functionality	TC-IO-001-01	PASS

Table 1: Requirements Coverage Matrix

3 DETAILED TEST CASES

3.1 Test Case TC-IO-001-01: Reset Test

Requirement: TR-IO-05
Purpose: Verify reset functionality and initial state.
Status: PASS

3.2 Test Case TC-IO-001-02: Write to Port A

Requirement: TR-IO-01

Purpose: Verify write functionality to Port A.

Status: PASS

3.3 Test Case TC-IO-001-03: Data Retention

Requirement: TR-IO-02, TR-IO-04

Purpose: Verify data retention when write is disabled.

Status: PASS

3.4 Test Case TC-IO-001-04: Read from Port B

Requirement: TR-IO-03

Purpose: Verify read functionality from Port B.

Status: PASS

3.5 Test Case TC-IO-001-05: Address Isolation

Requirement: TR-IO-05

Purpose: Verify address isolation for unsupported addresses.

Status: PASS

4 SOURCE CODES

4.1 IO_Unit.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity IO_Unit is
    Port (
        clk          : in  STD_LOGIC;
        reset        : in  STD_LOGIC;

        -- Control from Control Unit
        io_we         : in  STD_LOGIC;

        -- Connections with CPU
        io_addr_in    : in  STD_LOGIC_VECTOR (7 downto 0);
        io_data_out    : in  STD_LOGIC_VECTOR (7 downto 0);
        io_data_in    : out STD_LOGIC_VECTOR (7 downto 0);

        -- Physical pins of the processor
        port_out_a    : out STD_LOGIC_VECTOR (7 downto 0);
        pin_in_b      : in  STD_LOGIC_VECTOR (7 downto 0)
    );
end IO_Unit;

architecture Behavioral of IO_Unit is
    signal reg_port_a : std_logic_vector(7 downto 0);

begin
    process(clk, reset)
    begin
```

```

        if reset = '1' then
            reg_port_a <= (others => '0');

        elsif rising_edge(clk) then
            if io_we = '1' then
                case io_addr_in is
                    when x"01" =>
                        reg_port_a <= io_data_out;
                    when others =>
                        null;
                end case;
            end if;
        end if;
    end if;
end process;

port_out_a <= reg_port_a;

process(io_addr_in, pin_in_b, reg_port_a)
begin
    case io_addr_in is
        when x"01" =>
            io_data_in <= reg_port_a;
        when x"02" =>
            io_data_in <= pin_in_b;
        when others =>
            io_data_in <= (others => '0');
    end case;
end process;

end Behavioral;

```

4.2 tb_IO_Unit.vhd

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity tb_IO_Unit is
end tb_IO_Unit;

architecture Behavioral of tb_IO_Unit is

    component IO_Unit is
        Port (
            clk          : in  STD_LOGIC;
            reset        : in  STD_LOGIC;
            io_we         : in  STD_LOGIC;
            io_addr_in   : in  STD_LOGIC_VECTOR (7 downto 0);
            io_data_out  : in  STD_LOGIC_VECTOR (7 downto 0);
            io_data_in   : out STD_LOGIC_VECTOR (7 downto 0);
            port_out_a   : out STD_LOGIC_VECTOR (7 downto 0);
            pin_in_b     : in  STD_LOGIC_VECTOR (7 downto 0)
        );
    end component;

    signal clk_tb          : STD_LOGIC := '0';
    signal reset_tb        : STD_LOGIC := '0';
    signal io_we_tb        : STD_LOGIC := '0';
    signal io_addr_in_tb   : STD_LOGIC_VECTOR (7 downto 0) := (others => '0');
    signal io_data_out_tb  : STD_LOGIC_VECTOR (7 downto 0) := (others => '0');
    signal io_data_in_tb   : STD_LOGIC_VECTOR (7 downto 0);
    signal port_out_a_tb   : STD_LOGIC_VECTOR (7 downto 0);

```

```

signal pin_in_b_tb      : STD_LOGIC_VECTOR (7 downto 0) := (others => '0');

constant CLK_PERIOD : time := 10 ns;
constant ADDR_PORT_A : STD_LOGIC_VECTOR(7 downto 0) := x"01";
constant ADDR_PORT_B : STD_LOGIC_VECTOR(7 downto 0) := x"02";

begin

  UUT: IO_Unit port map (
    clk      => clk_tb,
    reset    => reset_tb,
    io_we     => io_we_tb,
    io_addr_in => io_addr_in_tb,
    io_data_out => io_data_out_tb,
    io_data_in  => io_data_in_tb,
    port_out_a  => port_out_a_tb,
    pin_in_b    => pin_in_b_tb
  );

  clk_process : process
  begin
    clk_tb <= '0';
    wait for CLK_PERIOD/2;
    clk_tb <= '1';
    wait for CLK_PERIOD/2;
  end process;

  stim_proc: process
  begin
    report "START: Test Bench for IO_Unit (PolyFlop8)";

    reset_tb <= '1';
    io_we_tb <= '0';
    io_addr_in_tb <= x"00";
    io_data_out_tb <= x"00";
    pin_in_b_tb <= x"00";
    wait for 100 ns;

    report "TC-IO-01: Reset Test";
    reset_tb <= '0';
    wait for CLK_PERIOD;
    assert port_out_a_tb = x"00"
      report "Error TC-IO-01: Port A was not zeroed after Reset!" severity
        error;

    report "TC-IO-02: Write 0xAA to Port A (Address 0x01)";
    io_addr_in_tb <= ADDR_PORT_A;
    io_data_out_tb <= x"AA";
    io_we_tb <= '1';
    wait for CLK_PERIOD;
    io_we_tb <= '0';
    wait for 1 ns;
    assert port_out_a_tb = x"AA"
      report "Error TC-IO-02: Port A did not accept value 0xAA" severity
        error;

    report "TC-IO-03: Output Retention Test (Write Enable=0)";
    io_data_out_tb <= x"FF";
    io_addr_in_tb <= ADDR_PORT_A;
    wait for CLK_PERIOD;
    assert port_out_a_tb = x"AA"
      report "Error TC-IO-03: Data was overwritten despite no write signal
        error;

```

```

        report "TC-I0-04: Read from Port B (Address 0x02)";
        pin_in_b_tb <= x"55";
        io_addr_in_tb <= ADDR_PORT_B;
        io_we_tb      <= '0';
        wait for CLK_PERIOD;
        wait for 1 ns;
        assert io_data_in_tb = x"55"
            report "Error TC-I0-04: Incorrect read from physical pins of Port B"
                ↪ severity error;

        report "TC-I0-05: Attempt to write to unsupported address (0xFF)";
        io_addr_in_tb <= x"FF";
        io_data_out_tb <= x"FF";
        io_we_tb      <= '1';
        wait for CLK_PERIOD;
        io_we_tb      <= '0';
        assert port_out_a_tb = x"AA"
            report "Error TC-I0-05: Write to invalid address changed Port A"
                ↪ state!" severity error;

        report "END: All IO Unit tests completed.";
        wait;
    end process;
end Behavioral;

```

5 SIMULATION LOG

```

Time resolution is 1 ps
Note: START: Test Bench for IO_Unit (PolyFlop8)
Time: 0 ps Iteration: 0 Process: /tb_IO_Unit/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ IO_TB.vhd
Note: TC-I0-01: Reset Test
Time: 100 ns Iteration: 0 Process: /tb_IO_Unit/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ IO_TB.vhd
Note: TC-I0-02: Write 0xAA to Port A (Address 0x01)
Time: 110 ns Iteration: 0 Process: /tb_IO_Unit/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ IO_TB.vhd
Note: TC-I0-03: Output Retention Test (Write Enable = 0)
Time: 121 ns Iteration: 0 Process: /tb_IO_Unit/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ IO_TB.vhd
Note: TC-I0-04: Read from Port B (Address 0x02)
Time: 131 ns Iteration: 0 Process: /tb_IO_Unit/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ IO_TB.vhd
Note: TC-I0-05: Attempt to write to unsupported address (0xFF)
Time: 142 ns Iteration: 0 Process: /tb_IO_Unit/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ IO_TB.vhd
Note: END: All IO_Unit tests completed.
Time: 152 ns Iteration: 0 Process: /tb_IO_Unit/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ IO_TB.vhd

```

6 PASS/FAIL CRITERIA

6.1 Success Criteria

Criteria	Description
Functional	All test cases execute without assertion failures
Timing	No timing violations (IO Unit is synchronous)
Coverage	100% statement coverage, 100% branch coverage achieved
Consistency	Identical results across multiple simulation runs
Documentation	All test results properly documented and signed off

6.2 Failure Conditions

- Any single assertion failure during simulation
- Code coverage less than 100% for IO Unit behavioral code
- Timing violations or simulation warnings
- Inconsistent results between simulation runs
- Missing or incomplete test documentation

6.3 Coverage Goals

- **Statement Coverage:** 100% of IO Unit process statements executed
- **Branch Coverage:** 100% of case statement branches taken
- **Condition Coverage:** All write and read conditions tested
- **Toggle Coverage:** All signal bits toggled 0→1 and 1→0
- **FSM Coverage:** N/A (IO Unit is not a state machine)

7 SPECIAL TEST CONSIDERATIONS

7.1 Edge Cases

- Test asynchronous reset during data write
- Verify data integrity during hold
- Test boundary conditions for data input
- Verify correct behavior during clock edge transitions
- Test all possible multiplexer combinations

7.2 Timing Considerations

- IO Unit is synchronous for write and combinational for read
- Monitor for glitches or hazards during input transitions
- Verify stable outputs within specified propagation delay
- Test with minimum, typical, and maximum delay models

7.3 Integration Considerations

- Verify interface compatibility with Memory and Control Unit
- Test handshake signals with Control Unit
- Validate output format matches processor expectations
- Confirm timing compatibility with other processor components